

SE LAB 3

Project #27 – Warehouse Inventory & Pallet Location Tracker.

Architecture Selection & Justification

Layered Architecture

Architecture Selection

We chose Layered Architecture for the Warehouse Inventory & Pallet Location Tracker.

Architectural Choice

Layered Architecture is appropriate for the warehouse system because it separates the system into distinct levels of responsibility. The proposed architecture consists of an **Input/Presentation Layer, Business Logic Layer, and Data Layer**. This provides a clear structure for handling pallet identification, location management, rack weight validation, movement tracking, and warehouse data storage.

Reason 1 – Separation of Concerns

Layered Architecture separates different responsibilities into appropriate components. Pallet identification is handled separately from warehouse business operations such as location assignment and rack weight validation, while the Warehouse Database is responsible for data storage. This separation makes the system easier to understand, maintain, test, and modify.

Reason 2 – Maintainability and Simplicity

The warehouse system contains several related operations, including pallet identification, three-dimensional location management, weight validation, and movement tracking. Organizing these functions into logical layers reduces system complexity and makes changes easier to implement. A change in the data management component can be handled without redesigning the complete system.

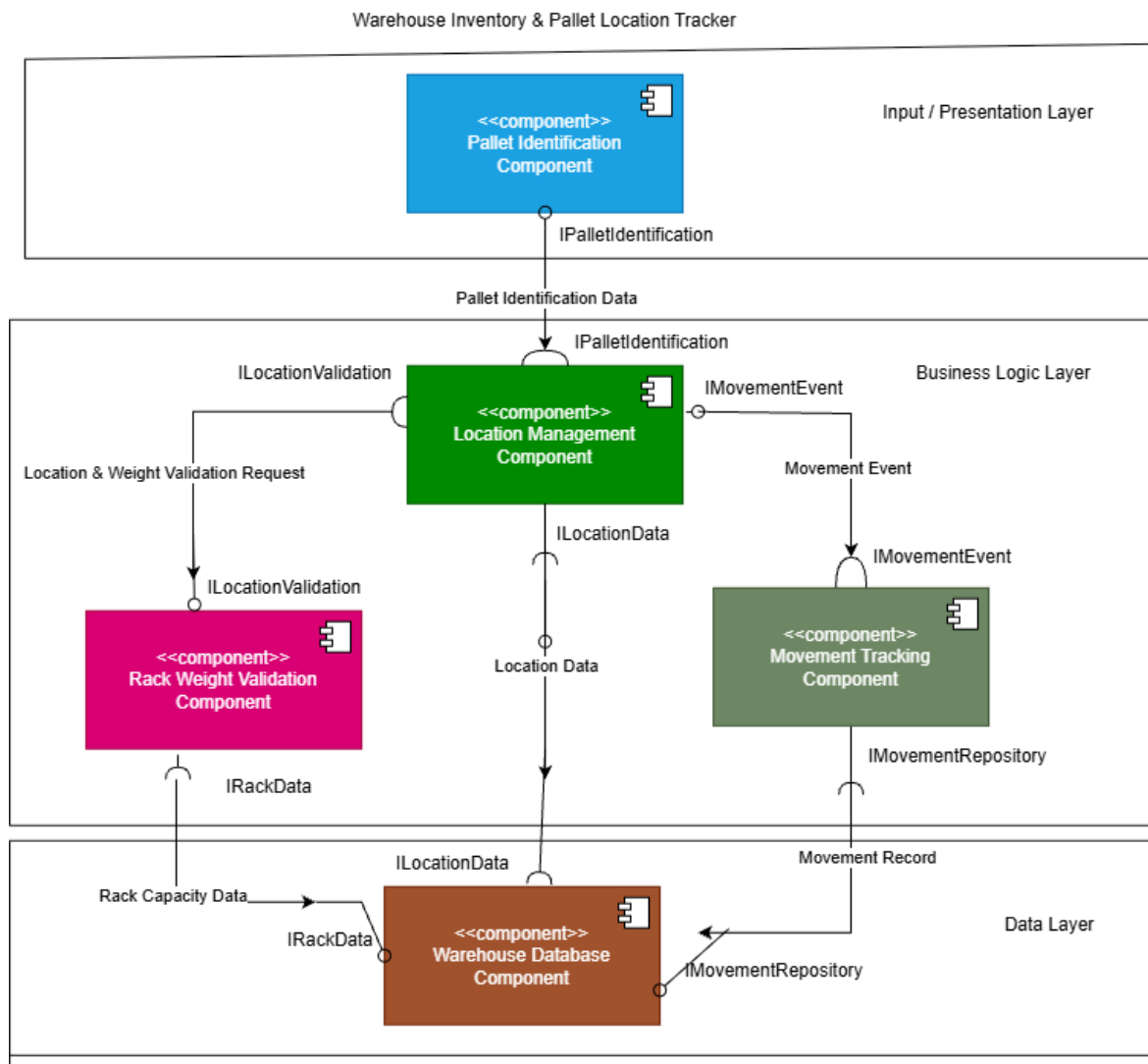
Security Advantage

Layered Architecture provides a security advantage by controlling access to the Warehouse Database through the Business Logic Layer. Users and scanning devices do not need direct access to sensitive inventory and movement data. Validation and authorization can be applied before information is stored, retrieved, or modified.

Performance Benefit

Layered Architecture can improve performance by keeping frequently used warehouse operations organized within dedicated components. Operations such as pallet identification and location lookup can be processed through the appropriate business components without unnecessary interaction with unrelated parts of the system. This supports responsive inventory and location operations.

UML COMPONENT DIAGRAM



UML Component Diagram – Layered Architecture

INPUT / PRESENTATION LAYER:

Pallet Identification Component

BUSINESS LOGIC LAYER:

Location Management Component

Rack Weight Validation Component

Movement Tracking Component

DATA LAYER:

Warehouse Database Component

INTERFACES:

1. IPalletIdentification

Pallet Identification Data

Pallet Identification Component → Location Management Component

2. ILocationValidation

Location & Weight Validation Request

Location Management Component → Rack Weight Validation Component

3. IRackData

Rack Capacity Data

Rack Weight Validation Component → Warehouse Database Component

4. ILocationData

Location Data

Location Management Component → Warehouse Database Component

5. IMovementEvent

Movement Event

Location Management Component → Movement Tracking Component

6. IMovementRepository

Movement Record

Movement Tracking Component → Warehouse Database Component